

head of prototypes. Simulations also provide essential information about feasibility, margins and limitations through data like thermal simulations.

Once you are satisfied with your new board design or want to create a custom board, then you are ready for fabrication

PCB Fabrication

Contrary to general perceptions, there are quite a few DIY PCB Fabrication methods if you know your way around circuits. The commonly used ones are:

- ♦ **Etching:** Quite similar to developing a photo, etching is a simple and quick process that offers low accuracy prototypes when you quickly need them. The process can be undertaken in most makerspaces and requires a UV light, PCB Developer solution and Etching solution.
- ♦ **CNC Milling:** An expensive and rare alternative, this does offer extremely precise fabrication. Although some components still need to be placed manually.
- ♦ **Fab Houses:** Outsource the fabrication. Period. Most Fab houses are capable of handling more complex designs but charge accordingly, so be sure of your design before ordering your board.

3D Printing a.k.a Additive Manufacturing – The Maker's cheatcode

Concepts like printing 3-Dimensional objects were stuff science fiction novels used to be made of. Although the creation of 3D objects was well underway since the 80s using casting and cross sectional methods, it is only in the 2000s that additive manufacturing really matured and subtractive manufacturing gained traction. Pretty soon, 3D printing wasn't restricted to metal shaping related tasks and started to be used for Polymer, Inkjet and Stereolithography printing.

The first step in 3D printing is of course creating the 3D model. This is done using various CAD tools and is similar to sculpting. Another method that has emerged recently is scanning the 3D structure information of a real world object using a camera sensor and creating a design file out of it.

Once the design is ready it is checked for basic errors in construction (non connecting surfaces, gaps) and then fed to a “slicer” which then splits it into a series of cross sectional layers. This information is then fed to client-side software that instructs the actual printing process. Here, resolution refers